Phase-Shear MLE: Systematic-Bias Correction Results

1e6 vs. 1e8 Atoms, Fitted-Feature vs. Oracle Regression

1 Setup

For each shot, `phase_shear_fit.py` fits a Gaussian $\times$ fringe model directly to the ground-state port image (a direct image-space MLE, independent of the PSMAP-based `map_inference.py` pipeline) at both Z0 and Z100, giving a per-shot phase estimate ϕ_0^{fit} together with fitted cloud-shape parameters $(\mu_x, \mu_y, \sigma_x, \sigma_y, \kappa_x, \kappa_y, \gamma_x, \gamma_y)$. ϕ_0^{fit} is dominated by the random per-shot laser phase (std $\sim \pi$), but carries a smaller systematic offset that depends on the cloud's actual position/shape on the fringe pattern (a wavefront-averaging effect). The differential $\Delta\phi = \phi_0^{\text{fit,Z100}} - \phi_0^{\text{fit,Z0}}$ cancels the shared random laser phase (std drops from $\sim$ 1.7–1.9 rad per AI to $\sim$ 70 mrad differential) and isolates the signal $\delta\phi_i = A_s \sin(2\pi f i) + A_c \cos(2\pi f i)$, but the systematic does not fully cancel and must be regressed out.

Two datasets were run apples-to-apples: the `.._kappa3.14e+04_both` runs (linear phase gradient applied at both AIs) at 10^6 and 10^8 atoms/shot, 20 runs $\times$ 20 shots each, same true signal ($A_s = 0.0878$, $A_c = 0.0479$, $f = 0.3$). Sample size was kept small (20 shots/run) for fast turnaround; a degree-1 (linear) polynomial model was used for the systematic-bias regression for numerical stability at this sample size (a full 16-feature degree-2 model overfits badly at $N \approx 380$, driving out-of-sample R^2 negative).

Two correction variants are compared:

- **Fitted-feature**: regress $\Delta\phi$ on the *fitted* cloud-shear parameters (the practically available correction). Reported both in-sample (IS) and via leave-one-run-out cross-validation (OOS).
- **Oracle**: regress $\Delta\phi$ on the *true* simulated initial kinematics instead of the fitted parameters – i.e. the idealised case where the cloud's real position/width is known exactly, with zero measurement error. Evaluated in-sample only, since the question this answers is "how good is the correction if the true parameters are already known," not generalisation.

2 Results

Dataset	Estimator	Residual std [mrad]	β RMSE [mrad]
10^6 atoms	raw (uncorrected)	13.84	5.35
	fitted-feature (in-sample)	6.65	2.63
	fitted-feature (leave-one-run-out)	7.22	—
	oracle, true kinematics (in-sample)	7.07	2.64
10^8 atoms	raw (uncorrected)	13.51	5.35
	fitted-feature (in-sample)	1.18	0.46
	fitted-feature (leave-one-run-out)	1.85	—
	oracle, true kinematics (in-sample)	1.08	0.33

Table 1: Differential-phase residual std (std of $\hat{\delta\phi} - \delta\phi_{\text{true}}$, pooled over all shots $\times$ runs) and per-run (A_s, A_c) recovery RMSE, before and after systematic-bias correction, for the fitted-feature and oracle (true-kinematics) regressions. β RMSE is the joint RMS error over A_s, A_c across the 20 per-run signal fits; not defined for the leave-one-run-out row (only the differential-phase residual was cross-validated, not a full per-run refit).

3 Discussion

At 10^8 atoms the correction is far more effective (raw 13.5 mrad $\rightarrow$ 1.1–1.2 mrad, an order of magnitude) than at 10^6 atoms (13.8 mrad $\rightarrow$ 6.6–7.1 mrad, roughly $2\times$), consistent with the image-space MLE localising the cloud shape and hence the systematic far more precisely at higher photon count. In both regimes the fitted-feature and oracle corrections land within statistical noise of each other in-sample (1e6: 6.65 vs. 7.07 mrad; 1e8: 1.18 vs. 1.08 mrad) – indicating the fitted cloud-shape estimates are already close to noise-free proxies for the true kinematics at this signal-to-noise level, so there is little room left for an oracle to improve on the practically available correction. The fitted-feature correction's out-of-sample degradation (leave-one-run-out vs. in-sample: +0.6 mrad at 1e6, +0.7 mrad at 1e8) reflects the usual bias/variance cost of estimating the correction from finite data rather than a fixed model.